

R6 Friction A level only

Name: _____

Class: _____

Date: _____

Time: **229 minutes**

Marks: **188 marks**

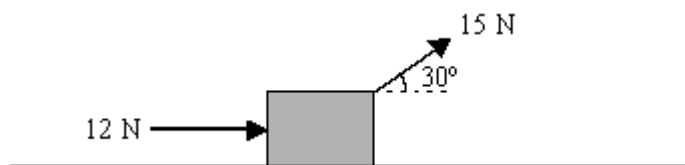
Comments:

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Q1.

The diagram shows a small box resting on a rough horizontal surface.



The box is of weight W newtons. It is pushed with a horizontal force of 12 newtons, and pulled with a force of 15 newtons at an angle of 30° to the horizontal.

The box rests in limiting equilibrium.

- (a) Draw a diagram to show all the forces acting on the box. (2)
- (b) Show that the frictional force acting on the box is approximately 25 newtons. (3)
- (c) The coefficient of friction between the box and the surface is $\frac{1}{3}$.
Find the normal reaction force between the box and the surface. (2)
- (d) Find the value of W . (3)

(Total 10 marks)

Q2.

A small case is moving on a rough horizontal surface. The case is being pushed with a force $(6\mathbf{i} - 2\mathbf{j})$ newtons, as shown in the diagram.



The case has mass 1.5 kg and moves with acceleration 0.6 m s^{-2} .

- (a) Draw a diagram to show the forces acting on the case. (1)
- (b) Show that the frictional force acting on the case is of magnitude 5.1 newtons. (3)
- (c) Find the normal reaction force between the case and the surface.

(3)

- (d) The coefficient of friction between the case and the surface is μ . Find the value of μ .

(2)

(Total 9 marks)

Q3.

A child travels down a slide at a **constant speed**. Model the slide as a rough plane inclined at an angle of 40° to the horizontal. Model the child as a particle of mass 20 kg and assume that there is no air resistance as the child moves.

- (a) (i) Draw a diagram to show the forces acting on the child.

(1)

- (ii) Show that the magnitude of the normal reaction force acting on the child is approximately 150 N.

(2)

- (iii) Find the magnitude of the friction force acting on the child and show that the coefficient of friction between the child and the slide is 0.84, correct to two significant figures.

(4)

- (b) In reality, air resistance acts on the child as she slides at a constant speed. How would this affect the value of the coefficient of friction that you calculated in part (a)(iii)?

(1)

(Total 8 marks)

Q4.

A particle, of mass 2 kg, is set into motion up a rough slope inclined at 40° to the horizontal. The coefficient of friction between the particle and the slope is 0.3. Assume that there is no air resistance acting on the particle.

- (a) Find the magnitude of the friction force that acts on the particle.

(3)

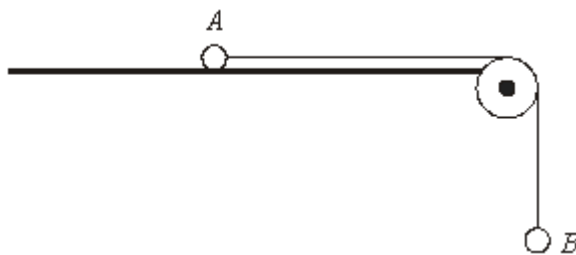
- (b) Find the acceleration of the particle as it moves up the slope.

(3)

(Total 6 marks)

Q5.

The diagram shows two particles, A and B , which are connected by a light inextensible string. The string passes over a smooth light pulley. Particle A has mass 20 kg and rests on a rough horizontal surface. The coefficient of friction between the particle and the surface is 0.15. Particle B has mass 5 kg and hangs with the string vertical, as shown in the diagram.

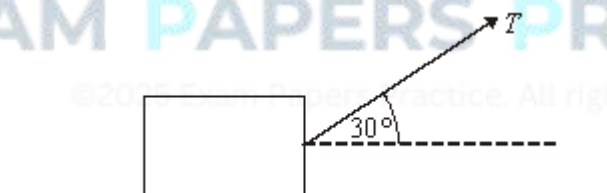


The particles are released from rest and begin to move.

- (a) Find the magnitude of the friction force acting on A . (2)
 - (b) By forming an equation of motion for each particle, show that the acceleration of each particle has magnitude 0.784 m s^{-2} . (5)
 - (c) Find the tension in the string. (2)
 - (d) Find the distance that A moves in the first 3 seconds of its motion. Assume that the acceleration of A remains constant during this time. (3)
- (Total 12 marks)

Q6.

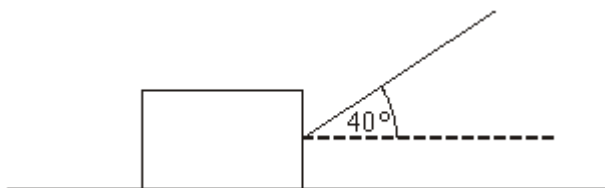
A heavy crate, of mass 200 kg , is pulled along a rough horizontal surface at a constant speed by a rope. The rope is at an angle of 30° to the horizontal. The tension in the rope is T newtons. The coefficient of friction between the crate and the surface is 0.6 . Model the crate as a particle.



- (a) Draw a diagram to show the forces acting on the crate. (1)
 - (b) Show that the magnitude of the normal reaction force on the crate is $(1960 - 0.5 T)$ newtons. (3)
 - (c) Find T . (5)
- (Total 9 marks)

Q7.

A box, of mass 50 kg, is dragged along rough horizontal ground by a rope attached to the box. The rope makes an angle of 40° with the ground and the tension in the rope is T newtons. The coefficient of friction between the box and the ground is 0.6. The diagram shows the box and the rope.



Model the box as a particle.

- (a) Draw a diagram to show the forces acting on the box. (1)
- (b) Find, in terms of T , the magnitude of the normal reaction force acting on the box. (3)
- (c) Show that the magnitude of the friction force acting on the box is $294 - 0.6T \sin 40^\circ$ (2)
- (d) The box accelerates at 0.5 m s^{-2} . Find T . (4)

(Total 10 marks)

Q8.

A rough plane is inclined at an angle of 40° to the horizontal. A particle, of mass 5 kg, is sliding down the plane.

- (a) Draw a diagram to show the forces acting on the particle. (1)
- (b) Find the magnitude of the normal reaction force acting on the particle. (2)
- (c) The coefficient of friction between the particle and the plane is 0.2. Show that the magnitude of the friction force acting on the particle is 7.51 N, correct to three significant figures. (2)
- (d) Show that the acceleration of the particle is 4.80 m s^{-2} , correct to three significant figures. (4)
- (e) Find the distance that the particle travels as its speed increases from 2 m s^{-1} to 10 m s^{-1} . (3)

(Total 12 marks)

Q9.

A crate, of mass 50 kg, is at rest on a warehouse floor. The floor is rough and horizontal. The coefficient of friction between the crate and the floor is μ . A rope is attached to the crate at an angle of 30° to the horizontal. The tension in the rope is 100 N. The crate is shown in the diagram.

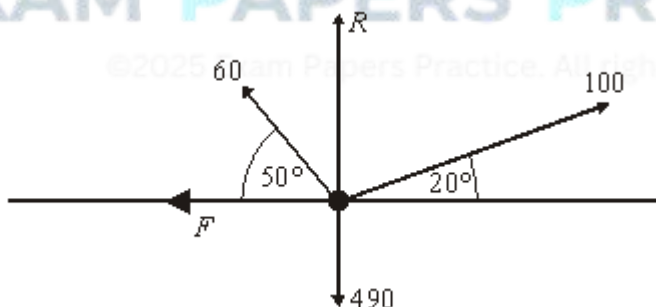


Model the crate as a particle.

- (a) Draw and label a diagram to show the forces acting on the crate. (1)
 - (b) Show that the magnitude of the normal reaction force acting on the crate is 440 N. (3)
 - (c) If the crate remains at rest, μ must satisfy the inequality $\mu \geq k$. Find k . (3)
 - (d) If $\mu = 0.1$, find the acceleration of the crate. (4)
- (Total 11 marks)**

Q10.

A box, of mass 50 kg, moves in a straight line on a rough horizontal surface. The diagram below shows **all** the forces acting on the box, as it moves. The magnitude of each force is in newtons. The box is modelled as a particle.



Note that the weight of the box has been included in the diagram.

- (a) Find R , the magnitude of the normal reaction force acting on the box. (3)
- (b) The box accelerates at 0.5 m s^{-2} .
 - (i) Show that F , the magnitude of the friction force acting on the box, is approximately 30.4.

(4)

- (ii) Find the coefficient of friction between the box and the surface.

(2)

(Total 9 marks)

Q11.

A box, of mass 20 kg, is initially at rest on a rough horizontal surface. A horizontal force of magnitude P newtons is applied to the box. The coefficient of friction between the box and the surface is 0.3.

- (a) State the magnitude of the normal reaction force acting on the box.

(1)

- (b) Find the magnitude of the friction force that acts on the box if:

(i) $P = 80$;

(ii) $P = 40$.

(3)

- (c) Find the value of P when the box is accelerating at 0.8 m s^{-2} .

(3)

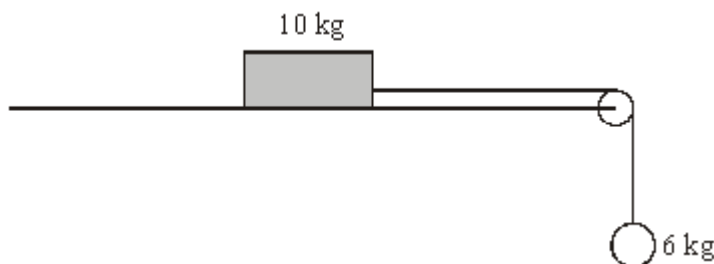
- (d) When the box reaches a speed of 6 m s^{-1} , the horizontal force P is removed. Find the distance that the box travels after the force P is removed.

(5)

(Total 12 marks)

Q12.

A block, of mass 10 kg, rests on a rough horizontal surface. It is connected by a light, inextensible string to a particle of mass 6 kg. The string passes over a light, smooth pulley, so that the string hangs vertically, as shown in the diagram.



Model the block as a particle.

- (a) The system is released from rest and the block travels 0.5 metres in 2 seconds. Find the acceleration of the system.

(2)

- (b) Show that the tension in the string is 57.3 N.

(3)

- (c) Find the coefficient of friction between the block and the plane.

(6)

(Total 11 marks)

Q13.

A particle of mass 60 kg is on a rough surface inclined at an angle of 40° to the horizontal.

- (a) Find the magnitude of the normal reaction force acting on the particle.

(2)

- (b) If the particle remains at rest, find the minimum value of the coefficient of friction between the particle and the slope.

(4)

- (c) If the coefficient of friction between the particle and the slope is 0.2, the particle slides down the slope. Find the acceleration of the particle in this case.

(5)

(Total 11 marks)

Q14.

A block of wood has mass 4 kg. It is placed on a rough horizontal surface and is pulled by a horizontal string. The coefficient of friction between the block and the surface is 0.4.

- (a) Draw a diagram to show the forces acting on the block.

(1)

- (b) Calculate the magnitude of the normal reaction force acting on the block.

(1)

- (c) If the acceleration of the block is 2 m s^{-2} , find the tension in the string.

(3)

- (d) If the tension in the string is 20 N, find the acceleration of the block.

(2)

(Total 7 marks)

Q15.

A block, of mass 5 kg, is held at rest on a rough plane, which is inclined at 30° to the horizontal. The block is released and slides down the plane. The coefficient of friction between the block and the plane is 0.2.

- (a) Draw a diagram to show the forces acting on the block as it slides.

(1)

- (b) Show that the magnitude of the friction force acting on the block is approximately 8.5 N.

(3)

- (c) Find the acceleration of the block. (4)
- (d) Find the speed of the block when it has travelled 1.2 metres down the slope. (3)
- (Total 11 marks)**

Q16.

A skier slides in a straight line directly down a slope inclined at 30° to the horizontal. The coefficient of friction between her skis and the slope is 0.3. The skier and her equipment are to be modelled as a particle of mass 80 kg. Assume that there is no air resistance present.

- (a) Draw a diagram to show the forces acting on the skier. (1)
- (b) (i) Find the magnitude of the normal reaction force acting on the skier. (2)
- (ii) Show that the magnitude of the friction force acting on the skier is 204 N to three significant figures. (2)
- (c) Find the acceleration of the skier. (4)
- (Total 9 marks)**

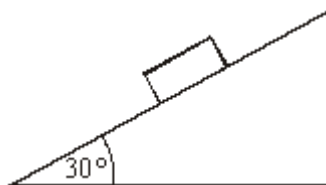
Q17.

A rope is used to pull a box, of mass 20 kg, up a rough slope. Assume that the box moves with a constant acceleration.

- (a) As the box moves 4 metres up the slope its speed increases from 1 m s^{-1} to 5 m s^{-1} . Calculate the acceleration of the box. (3)
- (b) The slope is at an angle of 30° to the horizontal. Assume that the rope is parallel to the slope. The coefficient of friction between the box and the slope is 0.4.
- (i) Draw a diagram to show the forces on the box. (1)
- (ii) Show that the magnitude of the friction force acting on the box is approximately 68 N. (3)
- (iii) Find the tension in the rope. (4)
- (Total 11 marks)**

Q18.

A block, of mass 7 kg, is placed on a rough slope that is inclined at 30° to the horizontal, as shown in the diagram. The block remains at rest in this position.

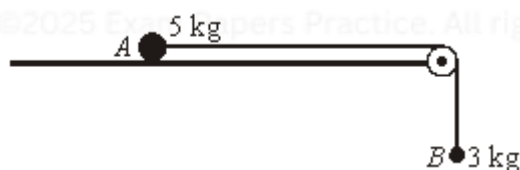


- (a) Draw a diagram to show the forces acting on the block. (1)
- (b) Find the magnitude of the normal reaction force acting on the block. (2)
- (c) Find the magnitude of the friction force acting on the block. (2)
- (d) The coefficient of friction between the block and the plane is μ . Find an inequality that μ must satisfy. (2)
- (e) A similar block, of mass 14 kg, is placed on the slope. Does this block remain at rest or slide? Give a reason for your answer. (2)

(Total 9 marks)

Q19.

The diagram shows two particles that are connected by a light inextensible string. The 5 kg particle, A , is on a rough horizontal surface and the 3 kg particle, B , hangs on the other end of the string as shown in the diagram. The string passes over a smooth light pulley.



The coefficient of friction between A and the horizontal surface is 0.8. The particles are set in motion with the string taut and A moving towards the pulley. The initial speed of both particles is 2 m s^{-1} . The particles slow down and stop before A reaches the pulley or B hits the floor.

- (a) Show that, while both particles are moving, the magnitude of their acceleration is 1.225 m s^{-2} . (6)
- (b) Find the tension in the string while the particles are moving. (2)

- (c) Find the distance that the particles move before they come to rest.

(3)

(Total 11 marks)



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